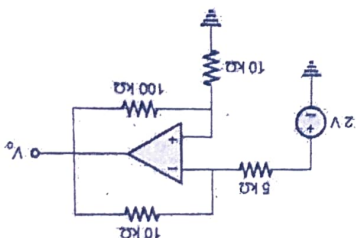
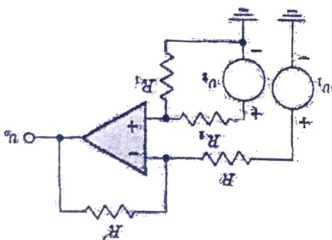


Note: Attempt all

1. Assume the op-amps in the following circuit are "ideal" as shown in the figure. Determine the output voltage (V_0).

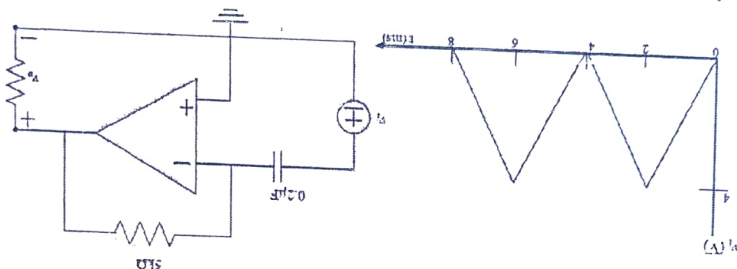


2. The circuit shown is a differential amplifier using an ideal OPAMP. Find the output voltage V_0 . Show that the output corresponding to the common mode voltage $V_c = \frac{1}{2}(V_1 + V_2)$ is equal to zero if $\frac{R'}{R} = \frac{R_2}{R_1}$. Find V_0 in this case. Also find the common mode rejection ratio of the amplifier if $\frac{R'}{R} \neq \frac{R_2}{R_1}$.



OR

- Sketch the output voltage waveform $V_0(t)$ for the circuit shown below for the given input voltage waveform.



3. Enlist various linear and non-linear applications of an OPAMP. Describe any two of them in detail alongwith schematics and working.